

Output tables for the test of Multiple comparisons.

April 29, 2025

1 Average rankings of Friedman test

Average ranks obtained by applying the Friedman procedure

Algorithm	Ranking
GWO	1
SA	2.9646
HC	2.0354

Table 1: Average Rankings of the algorithms

Friedman statistic considering reduction performance (distributed according to chi-square with 2 degrees of freedom: 764.989899.
P-value computed by Friedman Test: 2.3139268279237513E-10.

2 Post hoc comparisons

Results achieved on post hoc comparisons for $\alpha = 0.05$, $\alpha = 0.10$ and adjusted p-values.

2.1 P-values for $\alpha = 0.05$

<i>i</i>	algorithms	$z = (R_0 - R_i)/SE$	<i>p</i>
3	GWO vs. SA	27.645026	0
2	GWO vs. HC	14.568716	0
1	SA vs. HC	13.076311	0

Table 2: P-values Table for $\alpha = 0.05$

2.2 P-values for $\alpha = 0.10$

i	algorithms	$z = (R_0 - R_i)/SE$	p
3	GWO vs. SA	27.645026	0
2	GWO vs. HC	14.568716	0
1	SA vs. HC	13.076311	0

Table 3: P-values Table for $\alpha = 0.10$

2.3 Adjusted p-values

i	hypothesis	unadjusted p
1	GWO vs .SA	0
2	GWO vs .HC	0
3	SA vs .HC	0

Table 4: Adjusted p -values